Secure IoT in the Era of Quantum Computers

Where Are the Bottlenecks? — Schöffel, Lauer, Rheinländer & Wehn, TU Kaiserslautern (Sensors 2022)

POST-QUANTUM CRYPTOGRAPHY

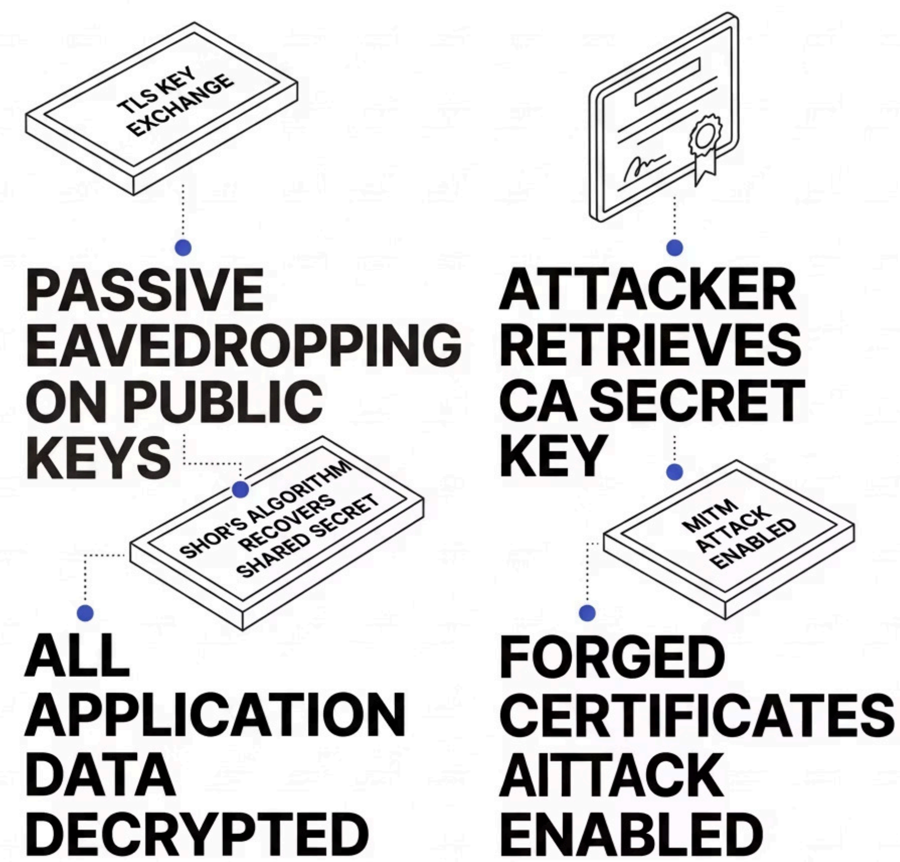
TLS · IOT · NIST ROUND 3

The Threat

Quantum Computers Will Break Today's PKC

Shor's algorithm solves integer factorisation and discrete logarithms in polynomial time — breaking RSA and ECDSA. Cryptographically relevant QCs are expected by **2031**.

The **"store now, decrypt later"** attack is already feasible: adversaries harvest encrypted traffic today for future decryption.



Made with GAMMA

NIST Standardisation

Post-Quantum Algorithm Families

Lattice-Based

KYBER, NTRU, SABER, Dilithium, Falcon — fast, compact. Primary NIST candidates.

Code-Based

Classic McEliece, BIKE, HQC — mature security, large keys. KEMs only.

Isogeny-Based

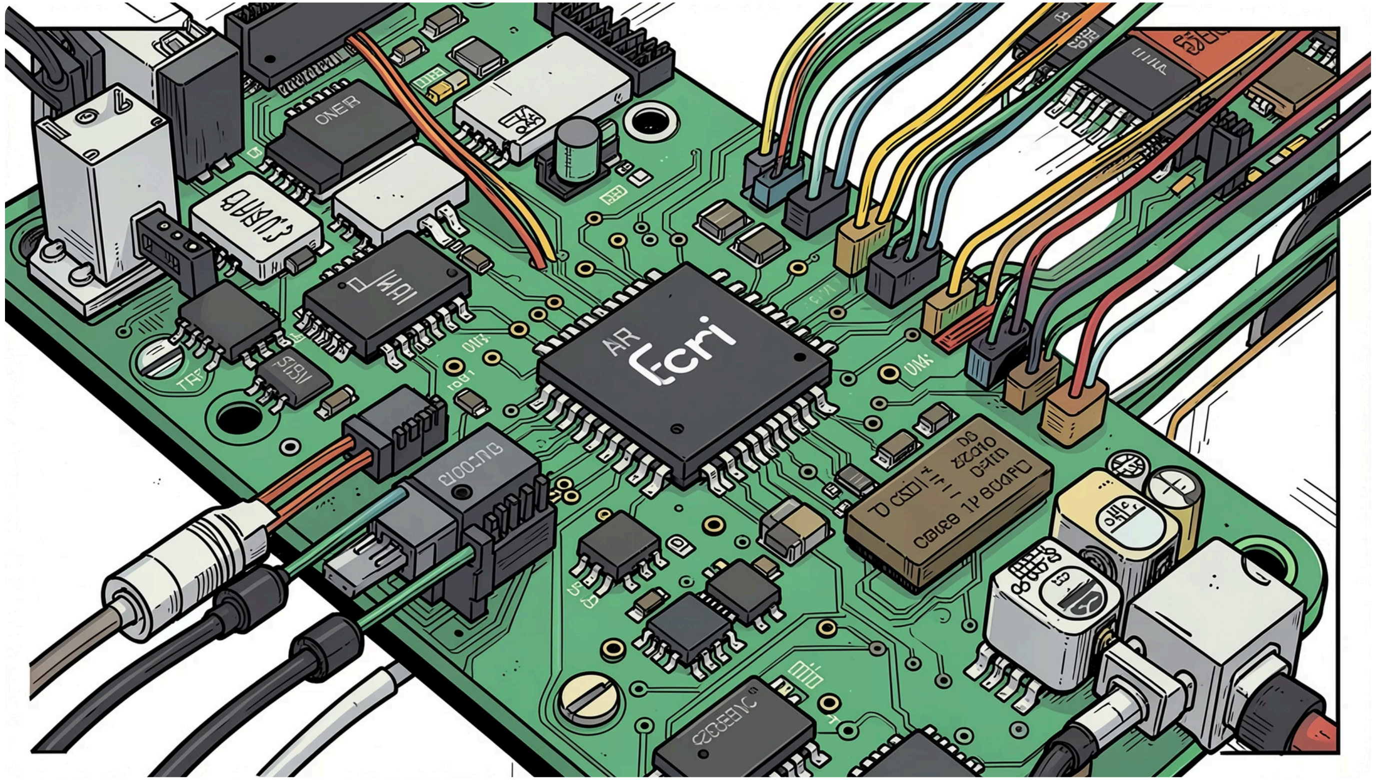
SIKE — smallest bandwidth, but computationally expensive.

Hash & Multivariate

SPHINCS+, Rainbow, GeMSS — tiny signatures or conservative design, but large public keys.

Evaluation Platform

A Representative Low-Power IoT Infrastructure



Edge Device — nRF52840 SoC

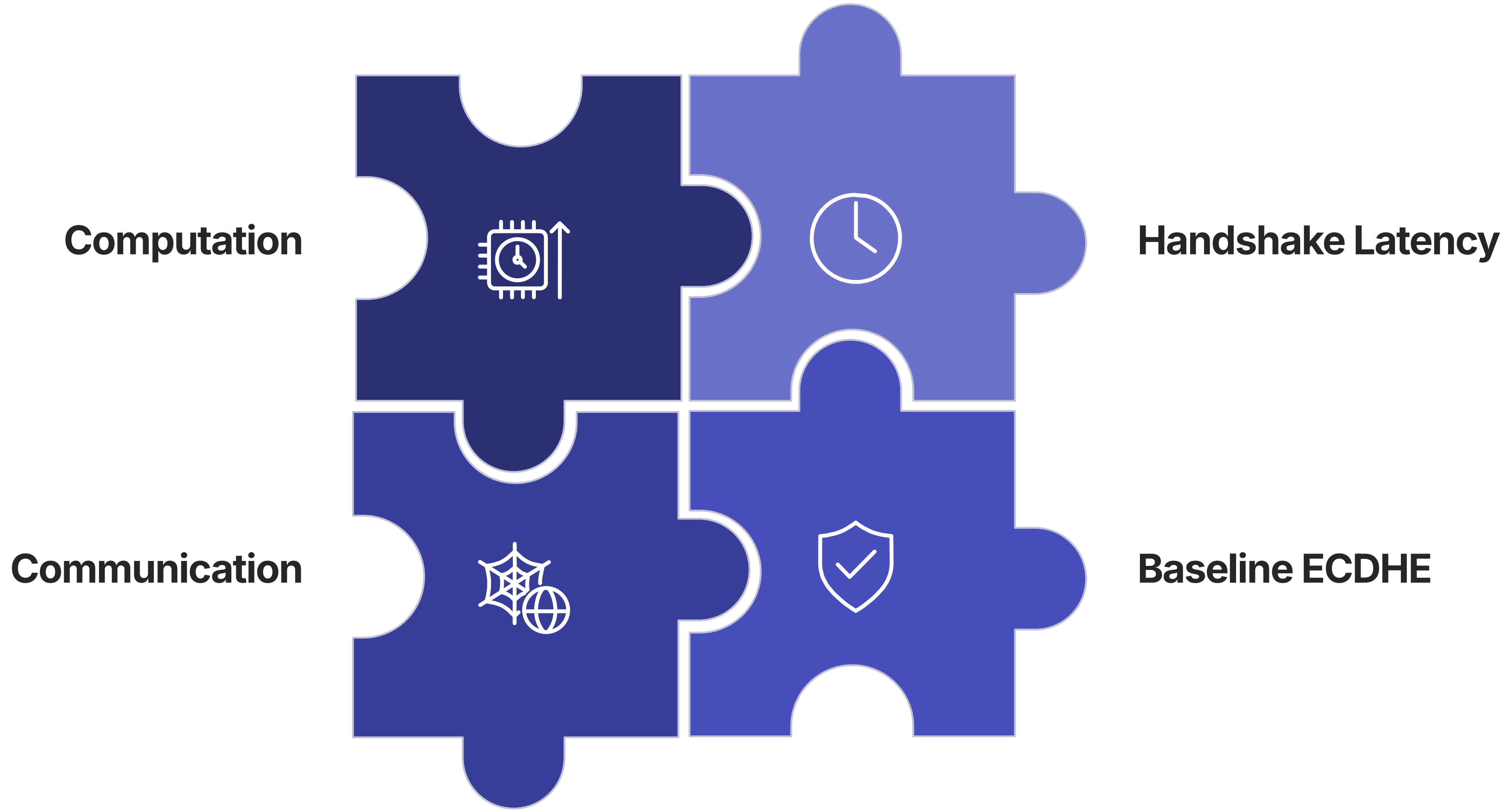
ARM Cortex-M4 MCU (NIST embedded reference platform), 256 KB RAM, 1 MB flash, ARM CryptoCell-310, Bluetooth 5 radio.

Protocol Stack

BLE → 6LoWPAN → TCP → TLS 1.2 → MQTT. Extended **mbedTLS** (client) and **OQS-OpenSSL** (server) with all NIST Round 3 KEMs and DSAs.

Key Finding #1

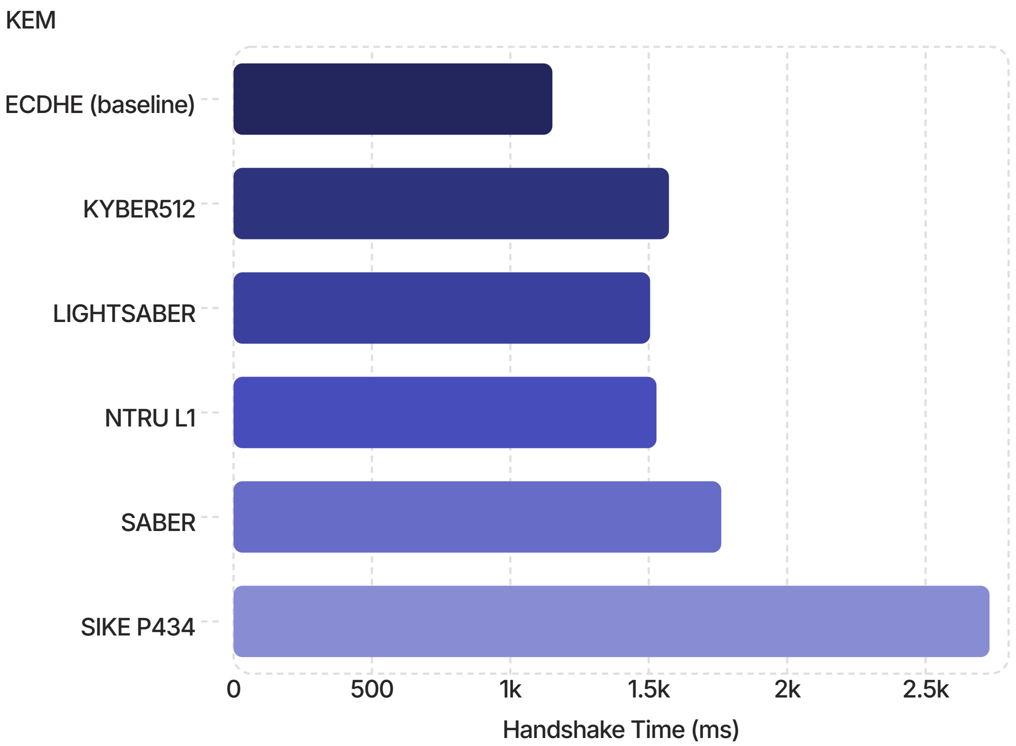
Bandwidth — Not Computation — Dominates Latency



For lattice-based finalists (KYBER, NTRU, SABER), encapsulation accounts for **under 5%** of total handshake time. Communication overhead from larger keys and ciphertexts is the primary bottleneck — fundamentally different from classical ECDHE, where software computation dominates.

KEM Results

Lattice-Based Finalists Are Ready for IoT



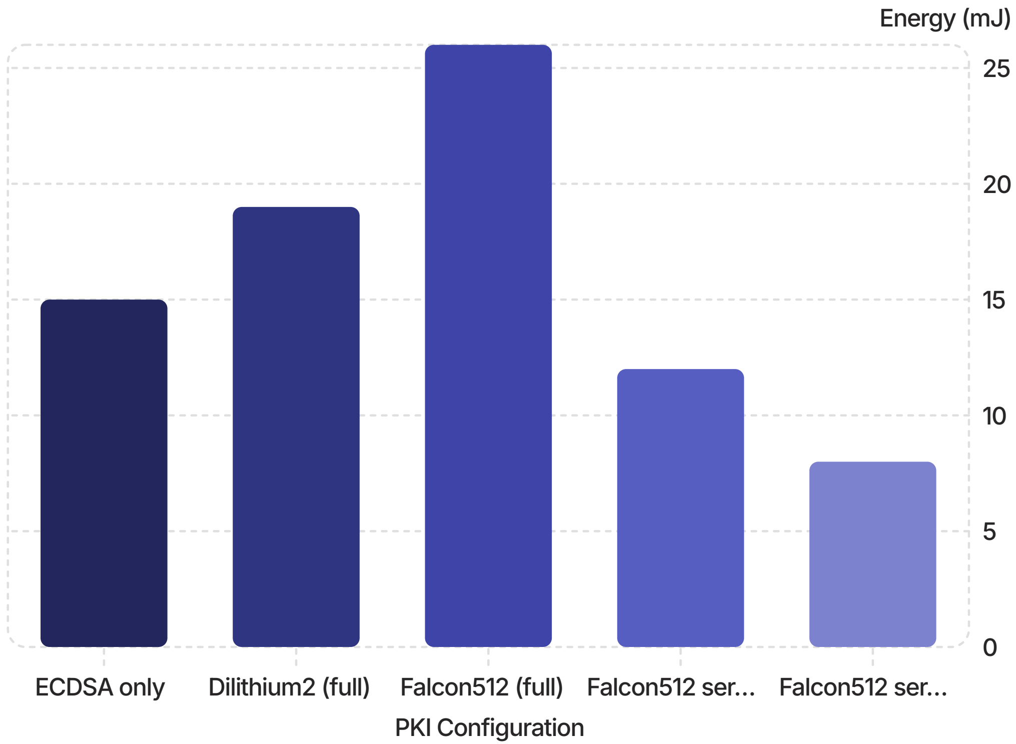
Key Observations

- Lattice finalists need ~25% more time than ECDHE at NIST Level 1
- SIKE demands **significantly more energy** due to complex isogeny calculations
- KYBER512 has the smallest stack footprint: **2.5 KB**
- Code-based KEMs (BIKE, HQC, FrodoKEM) have the largest memory requirements

Dedicated hardware accelerators offer very limited benefit for lattice-based KEMs — bandwidth, not computation, is the bottleneck.

DSA Challenge

Homogeneous PQ-DSAs Introduce Significant Overhead



The DSA Trade-Off

Dilithium2 — efficient computation, but large bandwidth. **Falcon512** — compact signatures, but computationally intensive signing on the client.

Homogeneous PQ-PKIs increase energy by up to **230%** and latency by **115%** versus ECDSA. A single Dilithium certificate chain can exceed TLS's 2^{14}-byte message limit.

Key Finding #2

Heterogeneous PKIs Are the Optimal Solution

🏆 Best Trade-Off

KYBER512 + Falcon512 (server/CA) + ECDSA (client)

Energy overhead: **30%** · Latency overhead: **50%**

Full PQ Protection

KYBER512 + Falcon512 (server/CA) + Dilithium2 (client)

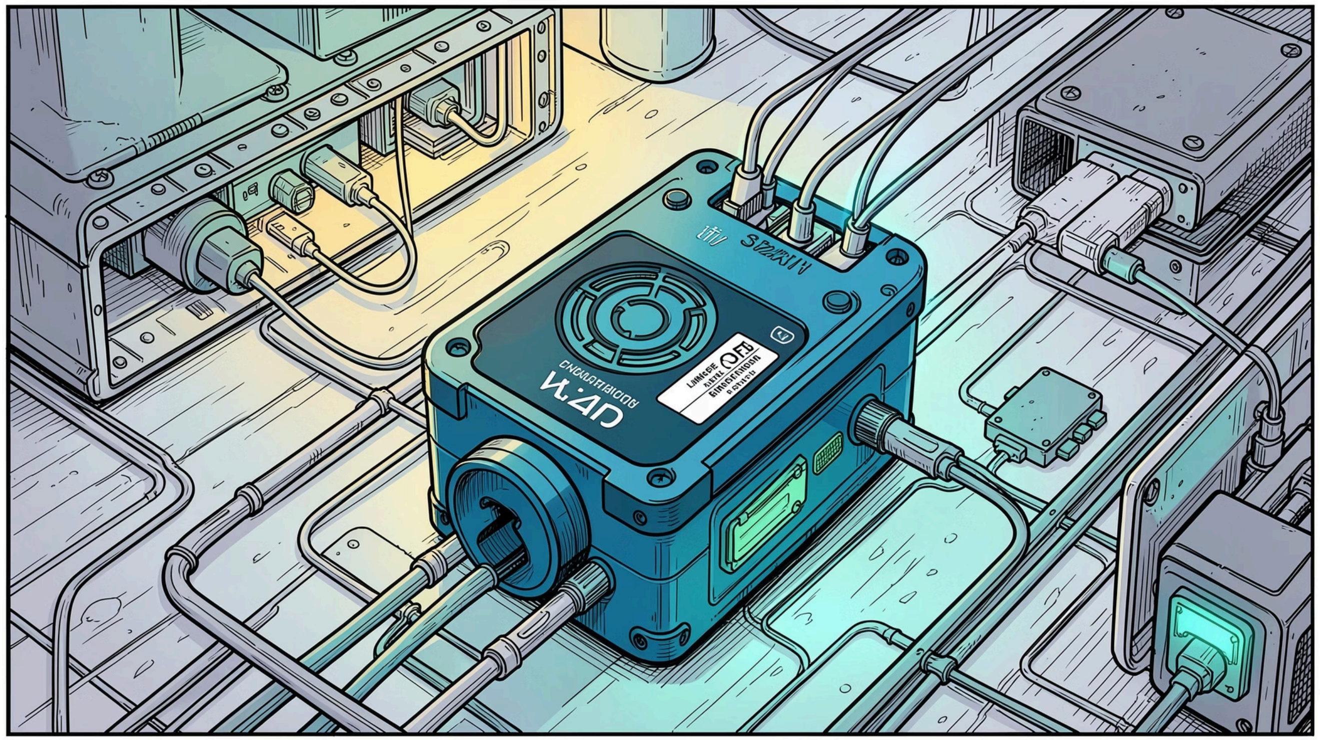
Energy reduced by 30% vs. homogeneous Falcon. Ideal where full quantum resistance is required.

NIST vs. Reality

NIST plans to standardise **one** DSA. This work demonstrates that combining algorithms by role yields superior efficiency.

Key Finding #3

PQC Is Feasible on Off-the-Shelf Hardware



Battery Life Analysis

Using a standard 1/2 AA Li-$SOCl_2$ cell (3.6 V, 1.2 Ah) with a representative MQTT sensor workload:

- KYBER512 + Falcon512/ECDSA PKI achieves **10+ years** with 200+ daily transfers
- Even fully Dilithium2-based PKIs maintain viable battery runtimes
- Migrating to PQC does **not significantly reduce** battery lifetime

This contradicts claims that dedicated hardware accelerators are mandatory for PQC on resource-constrained devices.

Conclusions

Five Key Takeaways

01

Holistic System View Is Essential

Evaluation must encompass the full protocol stack — not just cryptographic computation.

02

Bandwidth Is the Primary Bottleneck

Key and signature sizes drive overhead; hardware accelerators offer limited benefit for lattice-based schemes.

03

Heterogeneous PKIs Outperform Homogeneous Ones

Combining Falcon (server/CA) with ECDSA or Dilithium (client) and KYBER KEMs yields the best efficiency.

04

PQC Is Feasible Today

Off-the-shelf Cortex-M4 devices achieve viable battery runtimes with full PQC — no dedicated accelerators required.

05

Future Research Direction

Reducing signature and key sizes — not computation — is the most impactful path forward.